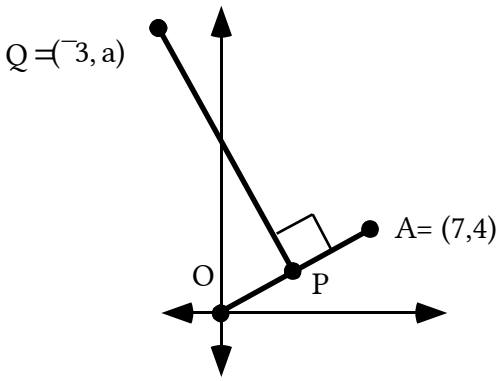
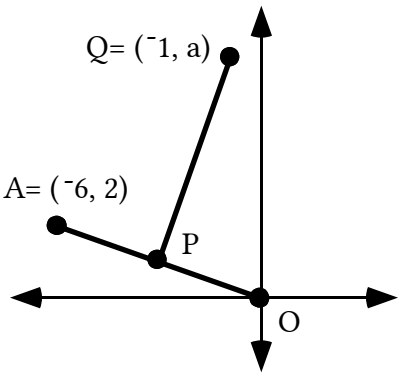
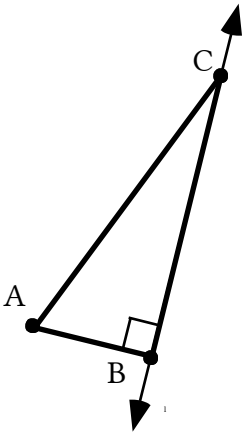
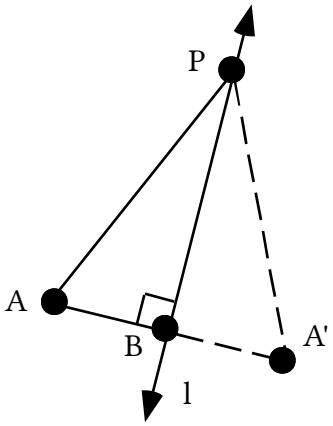
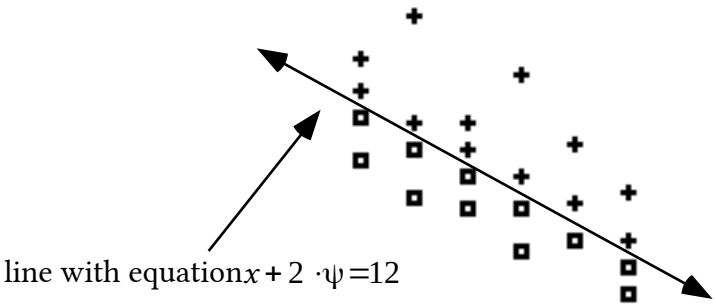

Q =(¯3, a)
A= (7,4)
O
P

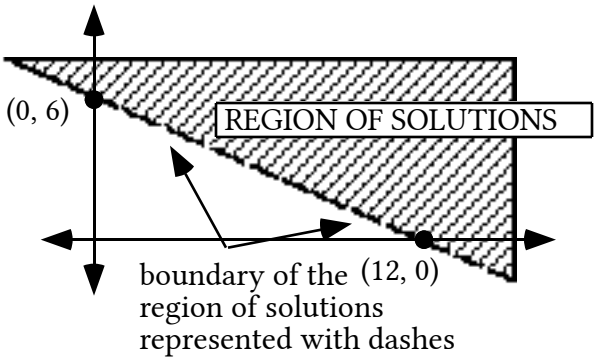

Q= (-1, a)
A= (-6, 2)
P
O


C
A
B

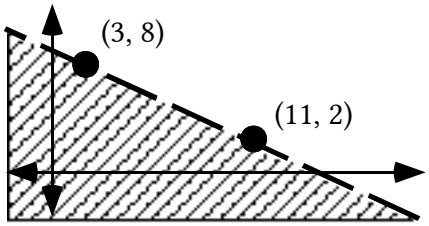
*	a	b	c	d	e	f	g
a	f	a	e	g	b	d	c
b	a	b	c	d	e	f	g
c	e	c	d	a	g	b	f
d	g	d	a	e	f	c	b
e	b	e	g	f	c	a	d
f	d	f	b	c	a	g	e
g	c	g	f	b	d	e	a

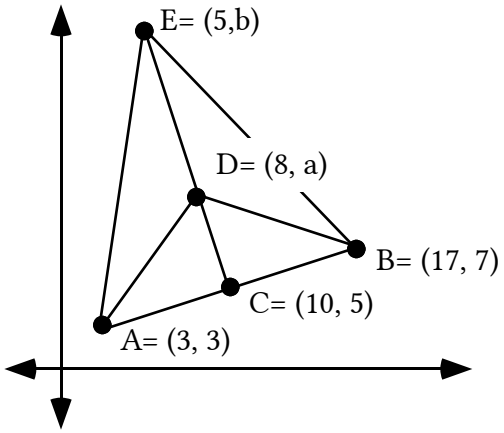

P
A
B
A'
l


line with equation $x + 2 \cdot \psi = 12$

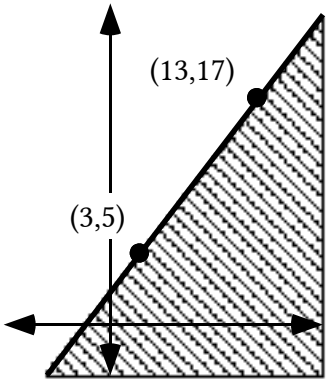


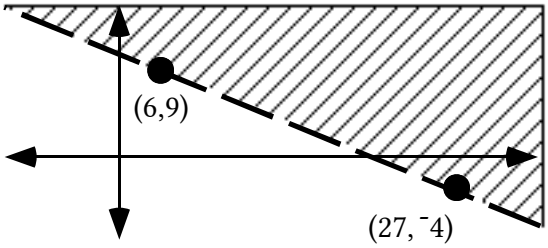
boundary of the region of solutions represented with dashes

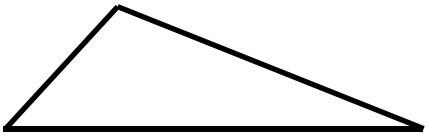

(3, 8)
(11, 2)


E= (5,b)
D= (8, a)
B= (17, 7)
C= (10, 5)
A= (3, 3)

element	$*$ -inverse of that element
a	
b	
c	
d	
e	
f	
g	

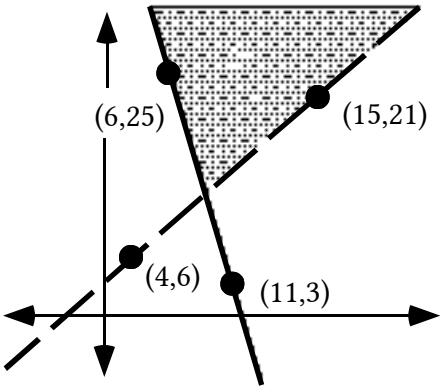

(13,17)
(3,5)

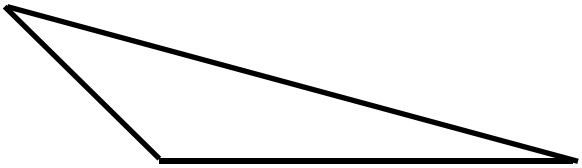

(6,9)
(27, -4)



*	a	b	c	d	e	f	g
a	f	a	e	g	b	d	c
b	a	b	c	d	e	f	g
c	e	c	d	a	g	b	f
d	g	d	a	e	f	c	b
e	b	e	g	f	c	a	d
f	d	f	b	c	a	g	e
g	c	g	f	b	d	e	a

$$\text{——}\circ(T_{(3,\,-7)} \circ X) = T_{(-3,7)} \circ T_{(6,3)}$$


(6,25)
(15,21)
(4,6)
(11,3)



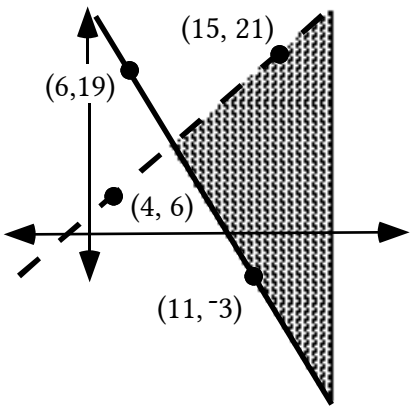
$$(R_{25} \circ X) = R_{40}$$

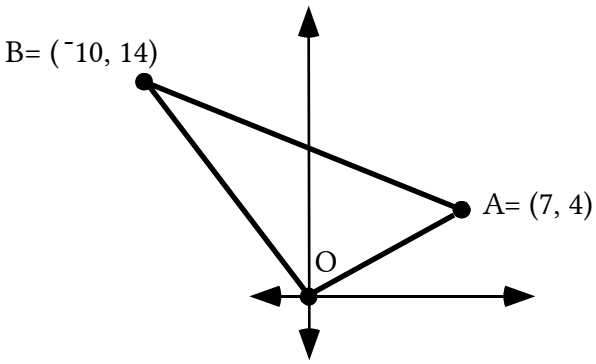
$R_{-25} \circ (R_{25} \circ X) = R_{-25} \circ R_{40}$ (attaching the inverse)

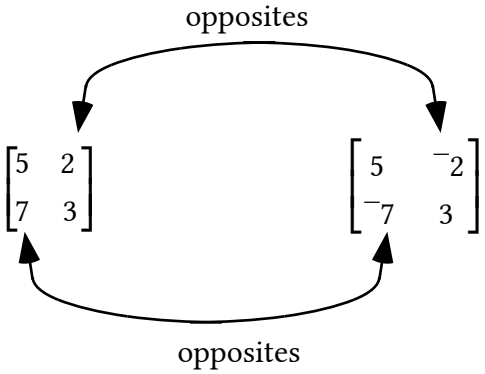
$(R_{-25} \circ R_{25}) \circ X = R_{15}$ (associating the inverses)

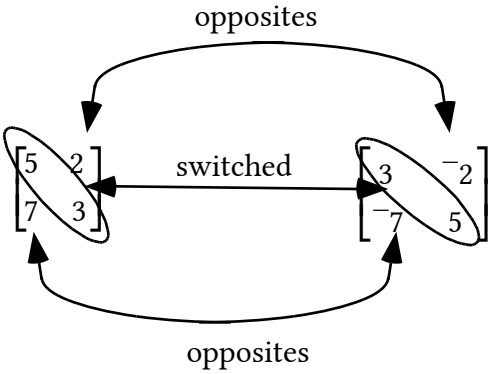
$R_0 \circ X = R_{15}$ (replacing associated inverses with the identity

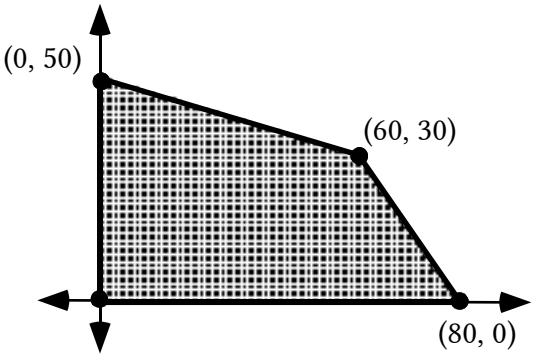
$X = R_{15}$ (letting identity “fade out”)

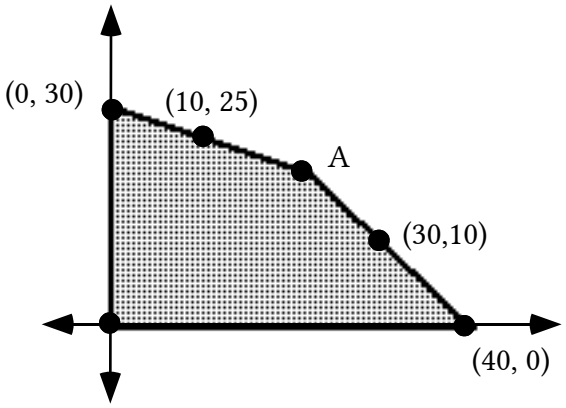

(15, 21)
(6,19)
(4, 6)
(11, -3)

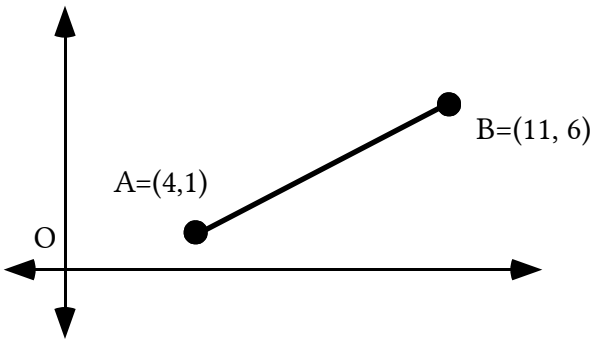

B= (¯10, 14)
A= (7, 4)
O

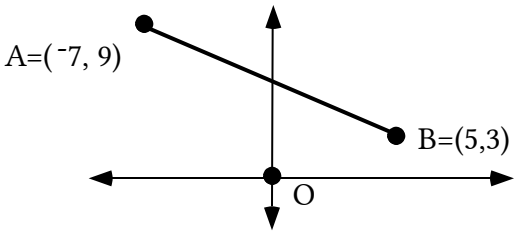

opposites
5 2
7 3
5 −2
−7 3
opposites

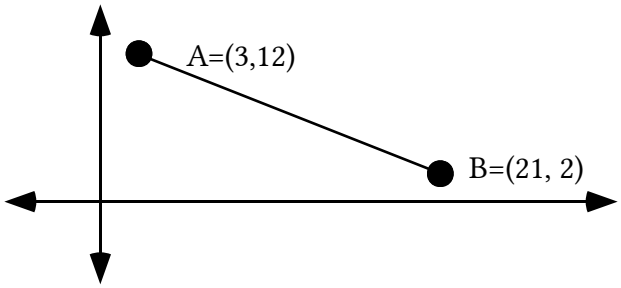

opposites
5
2
7
3
switched
3
−2
−7
5
opposites

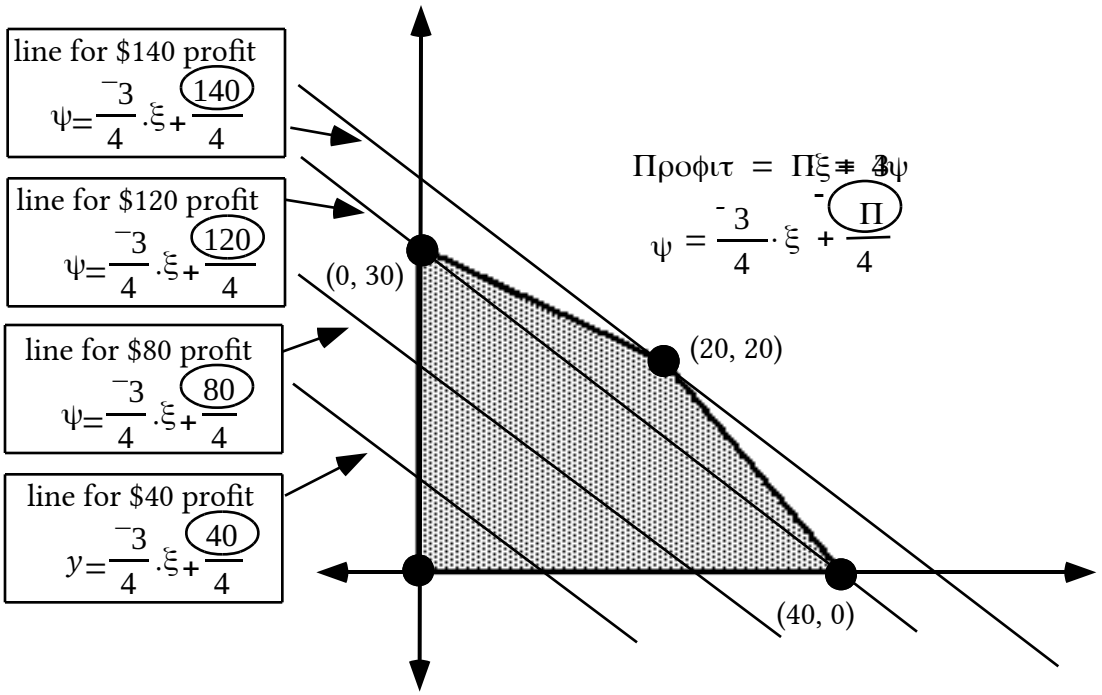

(0, 50)
(60, 30)
(80, 0)


(0, 30)
(10, 25)
A
(30,10)
(40, 0)


B=(11, 6)
A=(4,1)
O

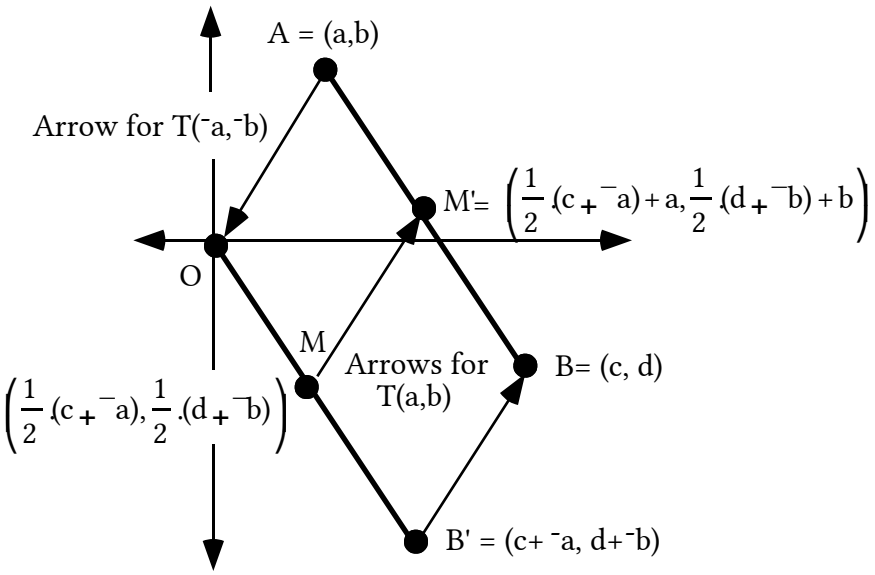

A=(-7, 9)
B=(5,3)
O

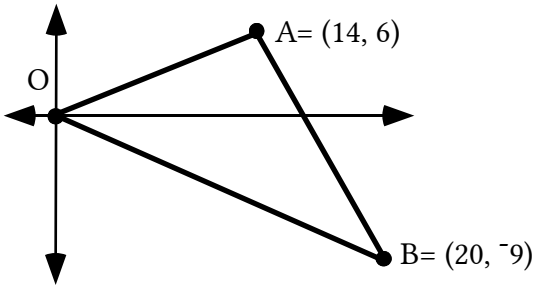

A=(3,12)
B=(21, 2)

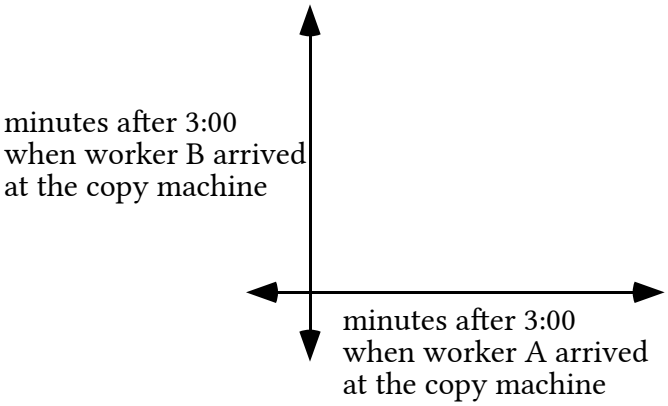

line for $140 profit
line for $120 profit
line for $80 profit
line for $40 profit
(0, 30)
(20, 20)
(40, 0)

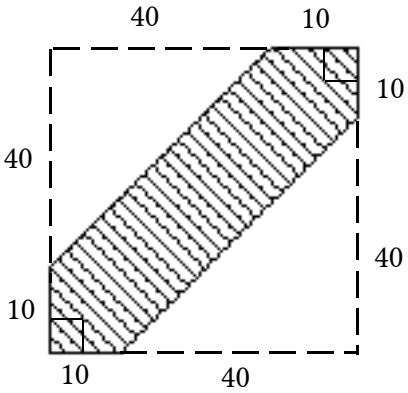
$$\frac{1}{2}\cdot(c+\bar{}a)+a=\frac{1}{2}\cdot c+(\ )\cdot a+a=\frac{1}{2}\cdot c+\bar{}\frac{1}{2}\cdot a+\frac{2}{2}\cdot a$$

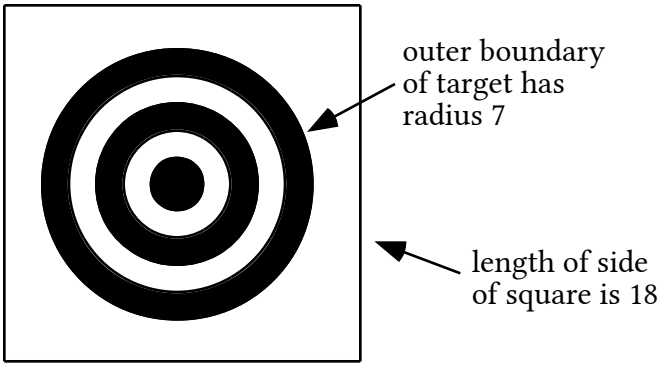
$$=\frac{1}{2}\cdot c+\left(\bar{}\frac{1}{2}+(\ )\right)\cdot a=\frac{1}{2}\cdot c+(\ )\cdot a=(\ )\cdot(a+c)$$

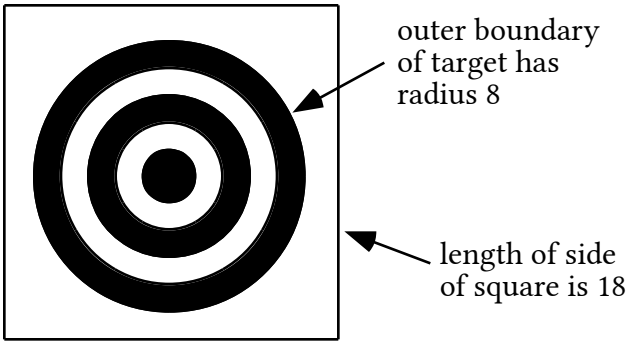

A = (a,b)
Arrow for T(-a,-b)
M'= $\left(\frac{1}{2}(c+-a)+a, \frac{1}{2}(d+-b)+b\right)$
O
M
Arrows for T(a,b)
B= (c, d)
$\left(\frac{1}{2}(c+-a), \frac{1}{2}(d+-b)\right)$
B' = (c+-a, d+-b)

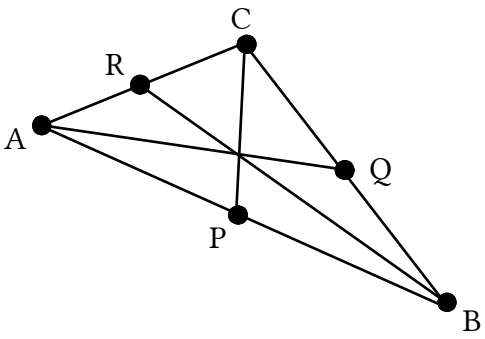

A= (14, 6)
O
B= (20, ¯9)

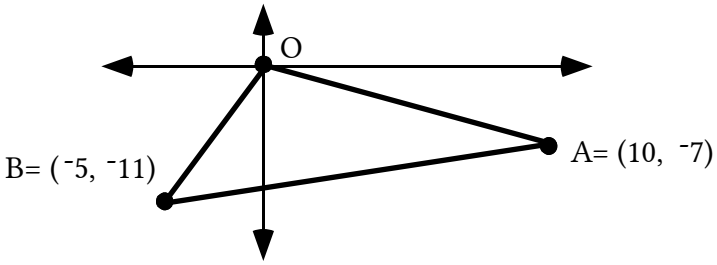

minutes after 3:00
when worker B arrived
at the copy machine
minutes after 3:00
when worker A arrived
at the copy machine

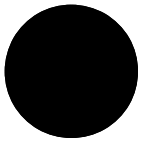

40
10
10
40
40
10
10
40


outer boundary
of target has
radius 7
length of side
of square is 18

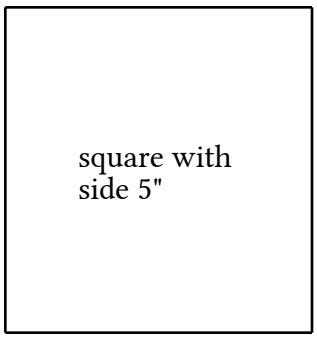

outer boundary
of target has
radius 8
length of side
of square is 18

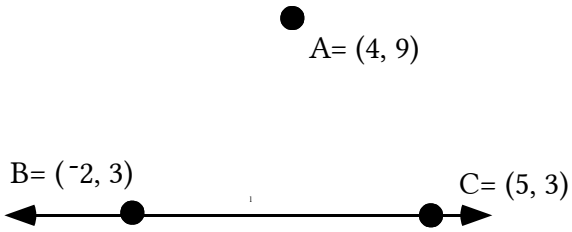

C
R
A
Q
P
B


O
B= (-5, -11)
A= (10, -7)

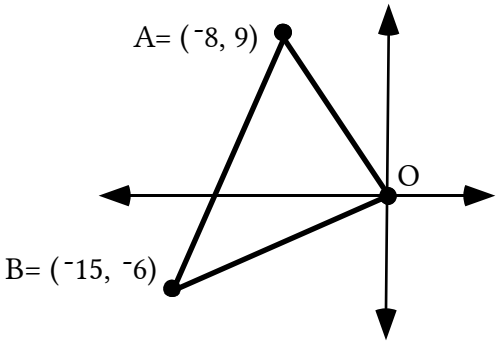


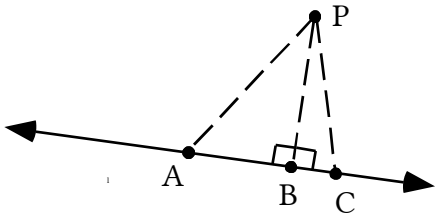
token with
diameter 2"

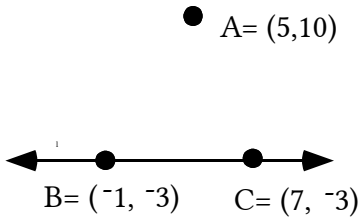


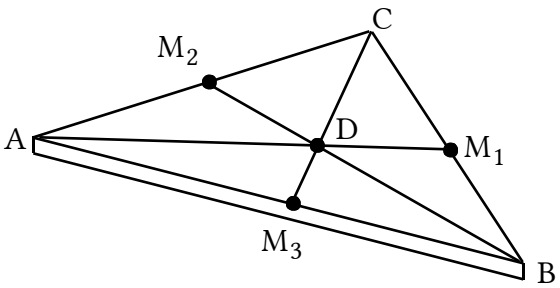

A= (4, 9)
B= (-2, 3)
C= (5, 3)

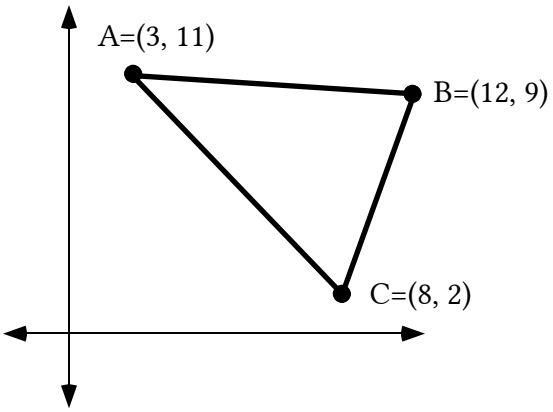
$$\begin{cases} x - y = \bar{}15 \\ y = 60 \end{cases}$$


A= (⁻8, 9)
O
B= (⁻15, ⁻6)


P
A
B
C


A= (5,10)
B= (-1, -3)
C= (7, -3)


C
M2
A
D
M1
M3
B


A=(3, 11)
B=(12, 9)
C=(8, 2)

$(3 \cdot x + 5)(4 \cdot x + \overline{\ }3) = 3 \cdot x \cdot (4 \cdot x + \overline{\ }3) + 5 \cdot$_____ =

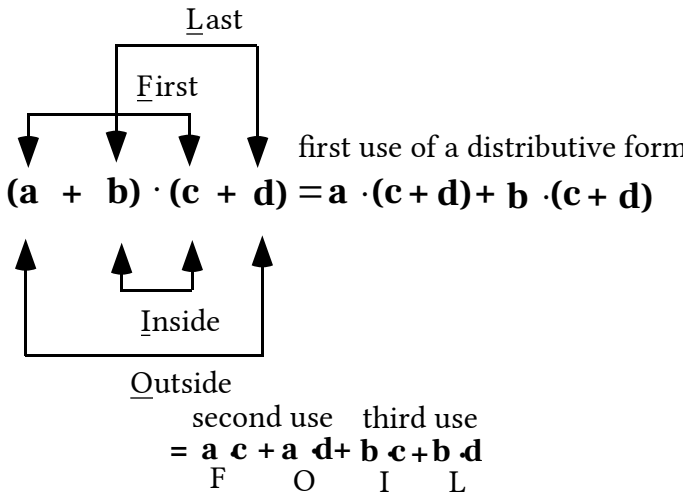
$3 \cdot x \cdot$____ $+ 3 \cdot x \cdot \overline{\ }3 + 5 \cdot 4 \cdot x + 5 \cdot$___ =

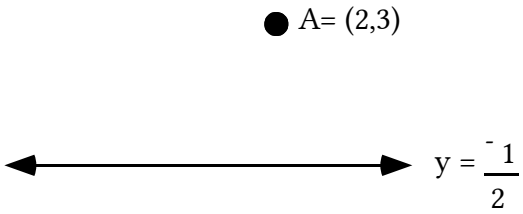
$12 \cdot$___ $+$ __ $\cdot x + 20 \cdot$ __ $+$ ___ =

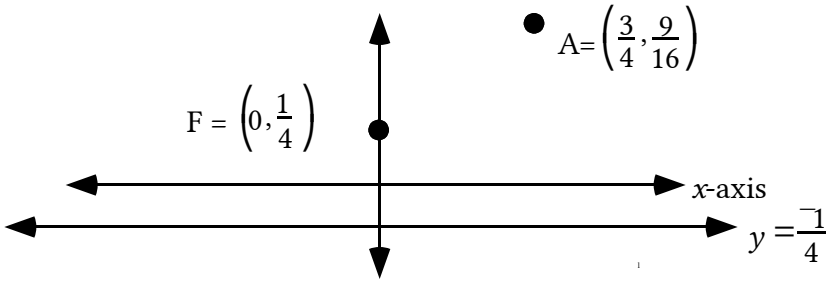
__ $\cdot x^2 +$ (___ $+$ ___) $\cdot x +$ ___ =

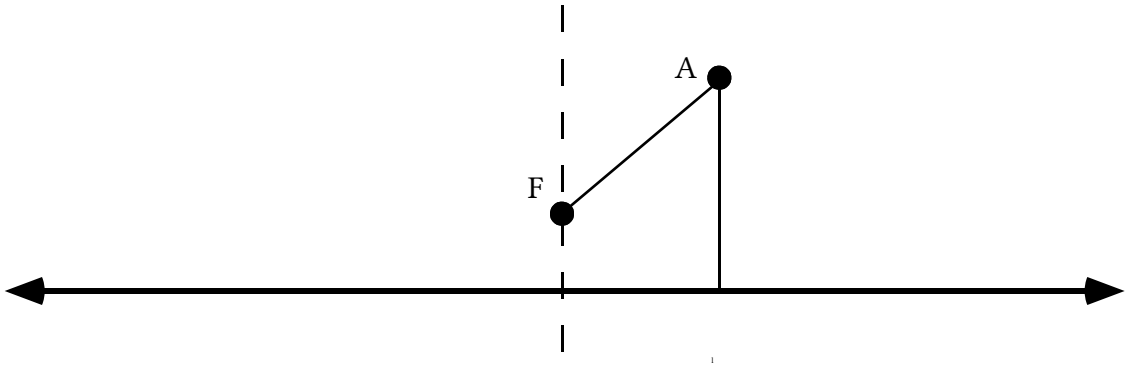
____ $+$ ____ $+$ ____

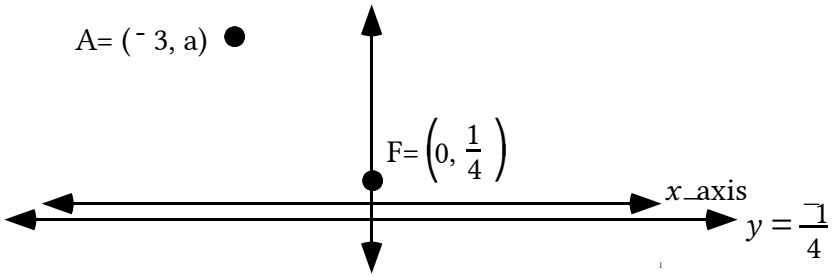
$$(3 \cdot 4,5 \cdot 16)$$

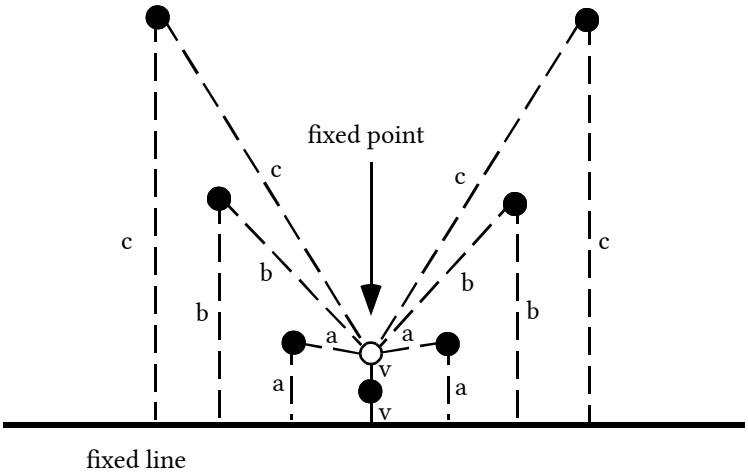

Last
First
first use of a distributive form
(a + b) · (c + d) = a · (c + d) + b · (c + d)
Inside
Outside
second use
third use
= a · c + a · d + b · c + b · d
F
O
I
L


A= (2,3)
y = - 1
2


A=$\left(\frac{3}{4},\frac{9}{16}\right)$
F = $\left(0,\frac{1}{4}\right)$
x-axis
$y=\frac{-1}{4}$


A
F


A= (- 3, a)
F= $\left(0, \frac{1}{4}\right)$
x_axis
$y = \frac{-1}{4}$


fixed point
c
c
c
c
b
b
b
b
a
a
a
a
v
v
fixed line